

Windows Server 2019 Notes for Exam Preparation (Expanded)

I. Windows Server 2019

- **Definition:** A robust server operating system developed by Microsoft, released in October 2018. It's built on the strong foundation of Windows Server 2016.
- **Purpose:** Designed to run a wide range of server roles, services, and applications in enterprise environments, from small businesses to large data centers.
- **Key Design Principles:** Focuses on four pillars: Hybrid Cloud, Security, Application Platform, and Hyper-Converged Infrastructure (HCI).
- **Editions:** Primarily available in Standard and Datacenter editions, with Datacenter offering more advanced features like Shielded VMs and Storage Spaces Direct.
- **Installation Options:** Supports Server Core (minimal interface, command-line management) for reduced attack surface and resource usage, and Desktop Experience (full GUI) for ease of management.

II. Types of Servers

- **Definition:** Software or hardware designed to provide specific functionalities or data to client computers over a network.
- **File Server:**
 - **Function:** Centralized storage for user and application files.
 - **Benefits:** Easier data backup and recovery, centralized access control (permissions), improved collaboration.
 - **Windows Services:** Often leverages NTFS permissions, Distributed File System (DFS) for namespace management and replication, and File Server Resource Manager (FSRM) for quotas and file screening.
- **Print Server:**
 - **Function:** Manages network printers and processes print jobs from clients.
 - **Benefits:** Centralized print queue management, driver deployment, secure printing.
 - **Windows Services:** Utilizes the Print and Document Services role.
- **Web Server:**
 - **Function:** Hosts websites, web applications, and web services. Responds to HTTP/HTTPS requests from web browsers.
 - **Windows Services:** Internet Information Services (IIS) is Microsoft's web server software, fully integrated with Windows Server. Supports ASP.NET, PHP, and other web technologies.
- **Database Server:**
 - **Function:** Stores, retrieves, and manages structured data for applications.
 - **Benefits:** Data integrity, concurrent access, robust query capabilities.
 - **Windows Examples:** Often hosts Microsoft SQL Server, but can also run other database systems like Oracle or MySQL.
- **Mail Server:**
 - **Function:** Handles the sending, receiving, and storage of electronic mail.
 - **Benefits:** Centralized email management, spam filtering, calendaring, and contact management.
 - **Windows Examples:** Microsoft Exchange Server is the primary mail server application for Windows environments.
- **Domain Server:**
 - **Function:** A broad term, but most commonly refers to a server running Active Directory Domain Services (AD DS). This server manages user identities, security, and network resources within a Windows domain.
 - **Benefits:** Centralized authentication, authorization, and management for a large number of users and devices.

III. Features of Windows Server 2019

- **Security Enhancements:**
 - **Windows Defender Advanced Threat Protection (ATP):** Provides post-breach detection, automated investigation, and remediation capabilities.
 - **Shielded Virtual Machines (VMs):** Encrypted VMs using BitLocker that can run on untrusted hosts, protecting against malicious administrators or malware. Requires Host Guardian Service.
 - **Attack Surface Analyzer:** Identifies vulnerabilities introduced by software installations.
 - **HTTP/2:** Supports HTTP/2 for faster and more secure web connections on IIS.
 - **Just Enough Administration (JEA):** Limits administrative privileges to specific tasks.
 - **Windows Server Hardening:** Continued focus on reducing attack surface and improving default security configurations.
- **Hybrid Cloud Capabilities:**
 - **Storage Migration Service:** Facilitates easy migration of file servers from older Windows Server versions or Linux Samba servers to Azure or newer Windows Servers, minimizing downtime.
 - **System Insights:** Local predictive analytics capabilities, powered by machine learning, for monitoring server performance, security, and resource usage.
 - **Azure Network Adapter:** Seamlessly connects on-premise Windows Servers to Azure Virtual Networks, extending the local network into the cloud.
 - **Azure Stack HCI:** A fully integrated hardware and software solution allowing enterprises to run Azure services on-premises.
- **Application Platform Improvements:**
 - **Kubernetes Support:** Enhanced platform for orchestrating containers, enabling large-scale deployment and management of containerized applications.
 - **Linux Containers on Windows:** Ability to run Linux containers directly on Windows Server through the use of an optimized Linux kernel.
 - **Smaller Server Core Image:** Reduced disk footprint (up to 50%) for Server Core installations, leading to fewer patches and improved security.
 - **Container Networking Enhancements:** Improved network configuration for containers, including support for Calico.
- **Hyper-Converged Infrastructure (HCI):**
 - **Storage Spaces Direct (S2D):** A software-defined storage technology that pools local storage from multiple servers into a single, highly available, and scalable storage solution for VMs. Key for building HCI solutions.
 - **Network Controller:** A centralized, programmable management point that automates the configuration, monitoring, and troubleshooting of network infrastructure.
 - **Improved Failover Clustering:** Enhancements to clustering for higher availability and disaster recovery for virtualized workloads. Includes cluster sets for massive scalability.

IV. Active Directory (AD)

- **Definition:** Microsoft's proprietary directory service that centralizes network administration, security, and resource access within a Windows domain environment.
- **Primary Function:** Provides authentication (verifying user identity) and authorization (determining what resources a user can access).
- **Logical Structure:** Comprises objects (users, groups, computers, OUs), domains, trees, and forests.
- **Physical Structure:** Includes domain controllers (DCs) and sites (for network topology awareness).
- **Schema:** Defines the types of objects that can be stored in AD and their attributes.

V. Active Directory Domain Services (AD DS)

- **Definition:** The fundamental server role that installs and enables the Active Directory database and associated services on a Windows Server.
- **Core Functions:**
 - **Centralized User/Computer Management:** Create, modify, and delete user and computer accounts.
 - **Authentication:** Verifies user credentials during logon.
 - **Authorization:** Grants or denies access to resources based on security permissions.
 - **Group Policy Management:** Enforces security settings, software deployment, and system configurations across the domain.
 - **DNS Integration:** AD DS relies heavily on DNS for name resolution and locating domain controllers.
 - **Trust Relationships:** Establishes secure communication between domains and forests.

VI. Object in Active Directory

- **Definition:** A distinct, identifiable entity within the Active Directory database, representing a resource or a security principal.
- **Attributes:** Each object is defined by a set of characteristics or properties (e.g., a user object has attributes like sAMAccountName, displayName, mail, phoneNumber).
- **Types of Objects:**
 - **User Objects:** Represent individual users.
 - **Computer Objects:** Represent machines joined to the domain.
 - **Group Objects:** Collections of users, computers, or other groups, used to simplify permissions management.
 - **Printer Objects:** Represents shared network printers.
 - **Shared Folder Objects:** Represents shared folders on the network.
 - **Organizational Units (OUs):** Container objects for organizing other objects.

VII. Domain in Active Directory

- **Definition:** A logical grouping of Active Directory objects (users, computers, groups) that share a common security boundary, administration, and database.
- **Contiguous DNS Namespace:** All computers within a domain typically share the same DNS domain name (e.g., contoso.com).
- **Security Boundary:** Provides a security boundary where group policies and permissions are applied.
- **Centralized Authentication:** Users authenticate once to the domain controller and gain access to resources throughout the domain via single sign-on (SSO).

VIII. Domain Controller (DC)

- **Definition:** A server that has the Active Directory Domain Services role installed and stores a writable copy of the Active Directory database.
- **Key Roles:**
 - **Authentication Provider:** Processes user logon requests and authenticates identities.
 - **Policy Enforcement:** Applies Group Policy Objects (GPOs) to users and computers.
 - **Database Management:** Stores and replicates the AD database with other DCs.
 - **Resource Access Control:** Facilitates authorization for resources within the domain.
 - **Flexible Single Master Operations (FSMO) Roles:** Hold specific critical roles (e.g., Schema Master, Domain Naming Master, PDC Emulator, RID Master, Infrastructure Master) that are unique within a forest or domain.
 - **Redundancy:** Multiple DCs per domain provide fault tolerance and load balancing.

IX. Global Catalog (GC)

- **Definition:** A special type of Domain Controller that stores a partial, read-only replica of all objects in every domain within the Active Directory forest.
- **Purpose:**
 - **Forest-wide Object Search:** Allows users and applications to search for any object in the entire forest without knowing which domain it resides in.
 - **Universal Group Membership:** Stores the universal group memberships for all users in the forest, which is critical for user logon process.
 - **Exchange Server Integration:** Used by Exchange Server to resolve recipient addresses.
- **Placement:** Typically, a few DCs in each site are configured as Global Catalogs to provide local search capabilities and reduce network traffic.

X. Organizational Unit (OU)

- **Definition:** A container object within an Active Directory domain that serves as the smallest unit to which Group Policy settings can be applied or administrative permissions can be delegated.
- **Purpose:**
 - **Hierarchical Organization:** Provides a logical and hierarchical structure for organizing AD objects (users, groups, computers, other OUs).
 - **Delegation of Administration:** Allows specific administrative tasks to be delegated to non-administrator users or groups for subsets of objects (e.g., HR can manage only HR user accounts).
 - **Group Policy Application:** The primary level at which Group Policy Objects (GPOs) are linked to apply settings to specific groups of users or computers. GPOs applied to an OU affect all objects within that OU and its child OUs.

XI. Tree and Forest in AD

- **Tree:**
 - **Definition:** A collection of one or more Active Directory domains that share a contiguous DNS namespace (e.g., company.com and sales.company.com).
 - **Trust Relationship:** Domains within a tree automatically establish two-way, transitive trust relationships. This means users in one domain can access resources in another domain within the same tree (if permissions allow).
 - **Shared Schema:** All domains in a tree share the same Active Directory schema.
- **Forest:**
 - **Definition:** The highest logical boundary in Active Directory. A collection of one or more Active Directory trees that share a common schema, configuration, and Global Catalog.
 - **Trust Relationship:** Trees within a forest automatically establish two-way, transitive trusts between their root domains.
 - **Common Schema:** All domains within the entire forest share a single, common Active Directory schema definition.
 - **Security Boundary:** While domains are security boundaries, the forest is a trust boundary. Any user with administrative control over the forest root

domain has control over the entire forest.

- **Purpose:** Allows organizations to manage multiple distinct DNS namespaces or separate business units while maintaining centralized management of the overall Active Directory infrastructure.

XII. DNS (Domain Name System)

- **Definition:** A fundamental hierarchical distributed naming system that translates human-readable domain names (e.g., www.example.com) into numerical IP addresses (e.g., 192.0.2.1) that computers use to identify services and devices on a network.
- **Critical for Active Directory:** AD DS heavily relies on DNS. Domain controllers register their services (like Kerberos, LDAP) in DNS, allowing clients to locate them for authentication and other services.
- **DNS Zones:** Store DNS records (e.g., A records for hostnames to IP, MX records for mail servers, SRV records for services).
- **Forward Lookup Zone:** Maps hostnames to IP addresses.
- **Reverse Lookup Zone:** Maps IP addresses to hostnames.

XIII. DHCP (Dynamic Host Configuration Protocol)

- **Definition:** A network management protocol used on IP networks for dynamically assigning IP addresses and other communication parameters (like subnet mask, default gateway, DNS server addresses) to devices connected to the network.
- **Benefits:**
 - **Automated Configuration:** Eliminates the need for manual IP configuration on each device.
 - **Reduced Administrative Overhead:** Simplifies network administration, especially in large networks.
 - **IP Address Management:** Prevents IP address conflicts.
 - **Mobility:** Allows devices to move between subnets and automatically acquire new IP configurations.
- **Process:** DHCP clients request configuration from a DHCP server, which then leases an IP address for a specified duration.
- **Windows Server Role:** The DHCP Server role is installed on Windows Server to provide this service.

XIV. File and Print Services

- **File Services:**
 - **Role:** Enables a server to function as a central repository for files that can be shared among network users, with robust access control.
 - **Key Components:**
 - **Shared Folders:** Network shares accessible by users.
 - **NTFS Permissions:** Granular security permissions applied directly to files and folders on NTFS-formatted volumes.
 - **Distributed File System (DFS):** Creates a unified namespace for shared folders located on different servers, simplifying access for users. DFS Namespace provides logical view, DFS Replication provides data synchronization.
 - **File Server Resource Manager (FSRM):** Tools for managing storage on file servers, including quotas (limiting space usage), file screening (preventing certain file types), and classification infrastructure.
 - **Work Folders:** Synchronizes user files from a corporate file server to users' devices.
- **Print Services:**
 - **Role:** Provides a centralized management point for network printers, print queues, and printer drivers.
 - **Key Components:**
 - **Print Server Role:** Installs the necessary components to manage printers.
 - **Print Queue Management:** Allows administrators to monitor, pause, restart, or delete print jobs.
 - **Driver Management:** Centralized repository for printer drivers, simplifying deployment to client machines.
 - **Secure Printing:** Features like PIN-release printing for confidential documents.
 - **Printer Pooling:** Allows multiple physical printers to share a single print queue for increased throughput.